

Presentación:

Nombre:

Algelis Severino

Tema:

Introducción a la Programación y Diagrama de flujo.

Tecnico:

Ciberseguridad Semipresencial

Facilitador:

Kevin Feliz Henriquez

Centro:

Diapiato Pro salud

Descripción:

En este documento .pdf adjunto capturas del reto 9 de 30 days of code de HackerRank.

Link GitHub:

<https://github.com/SeverinoCyber/30-days-of-code/tree/main/Hackerrank/day9>

Python3:

Problem

Objective

Recursive Method for Calculating Factorial

Function Description

Returns

Note:

Input Format

Constraints

Sample Input

Sample Output

Explanation

Today, we are learning about an algorithmic concept called recursion. Check out the [Tutorial](#) tab for learning materials and an instructional video.

$$factorial(N) = \begin{cases} 1 & N \leq 1 \\ N \times factorial(N - 1) & otherwise \end{cases}$$

Complete the factorial function in the editor below. Be sure to use recursion.

factorial has the following parameter:

- int n: an integer

- int: the factorial of n

Note: If you fail to use recursion or fail to name your recursive function factorial or Factorial, you will get a score of 0.

A single integer, n (the argument to pass to factorial).

- $2 \leq n \leq 12$
- Your submission must contain a recursive function named factorial.

3

6

Consider the following steps. After the recursive calls from step 1 to 3, results are accumulated from step 3 to 1.

Change Theme

Language Python 3

Exit Full Screen View

```
1 #!/bin/python3
2
3 import math
4 import os
5 import random
6 import re
7 import sys
8
9
10 # Complete the 'factorial' function below.
11 #
12 # The function is expected to return an INTEGER.
13 # The function accepts INTEGER n as parameter.
14 #
15
16 def factorial(n):
17
18     if n <= 1:
19         return 1
20
21     else:
22         return n * factorial(n - 1)
23
24 if __name__ == '__main__':
25     fptr = open(os.environ['OUTPUT_PATH'], 'w')
26
27     n = int(input().strip())
28
29     result = factorial(n)
30
31     fptr.write(str(result) + '\n')
32
33     fptr.close()
34
```

Line: 34 Col: 1

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Line: 34 Col: 1

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Test case 0

Compiler Message

Success

Test case 1

Test case 2

Input (stdin)

Expected Output

3

6

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Sample Input
3

Sample Output
6

Explanation
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Line: 38 Col: 6

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Test case 0

Compiler Message

Success

Test case 1

Input (stdin)

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Test case 2

Expected Output

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Change Theme Language C#

```
1 using System.CodeDom.Compiler;
2 using System.Collections.Generic;
3 using System.Collections;
4 using System.ComponentModel;
5 using System.Diagnostics.CodeAnalysis;
6 using System.Globalization;
7 using System.IO;
8 using System.Linq;
9 using System.Reflection;
10 using System.Runtime.Serialization;
11 using System.Text.RegularExpressions;
12 using System.Text;
13 using System;
14
15 class Result
16 {
17
18     /*
19      * Complete the 'factorial' function below.
20      *
21      * The function is expected to return an INTEGER.
22      * The function accepts INTEGER n as parameter.
23      */
24
25     public static int factorial(int n)
26     {
27         if(n==1){
28             return 1;
29         }else{
30             return n * factorial(n-1);
31         }
32     }
33
34 }
35
36
```

Line: 53 Col: 1

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